

**STELLENBOSCH UNIVERSITY
DEPARTMENT OF ECONOMICS**

ECONOMICS 114

MAIN ASSESSMENT A3: 21 JUNE 2022

Time: 3 hours

Marks: 100

EXAMINERS:

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Dr D Shepherd
Mr MP van Schoor

Ms L Boltman
Ms C Pengelly
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INTERNAL MODERATOR: Dr E Moses

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.

ANSWER **SECTION A** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS** NEXT TO THE NUMBER OF THE STATEMENT.

ALSO ANSWER **SECTION B** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS NEXT** TO THE NUMBER OF THE STATEMENT.

KEY:

AP = Average product
AFC = Average fixed cost
AVC = Average variable cost
AC = Average total cost
BL = Budget line
GDP = Gross domestic product
IC = indifference curve
K = capital
FF = Feasible frontier
MC = Marginal cost
MP = Marginal product
MPB = Marginal private benefit
MPC = Marginal private cost
MRS = Marginal rate of substitution

MRT = Marginal rate of transformation
MSB = Marginal social benefit
MSC = Marginal social cost
P = Price
PF = Production frontier
Q = Quantity
Q_d = Quantity demanded
Q_s = Quantity supplied
TC = Total cost
TP = Total product
w = Wage
WTA = Willingness to accept
WTP = Willingness to pay
Y = Income

SECTION A
TRUE/FALSE STATEMENTS

[10 marks: 1 mark for a correct answer and minus ½ for an incorrect answer]

1. GDP does not take the value of household or other types of unpaid work into account.
a) True
b) False
2. *Ceteris paribus* is an assumption in economic models which refers to keeping the important variables that you are focusing on, constant.
a) True
b) False
3. If Wendell increased his number of hours of free time following a decrease in his hourly wage, his income effect was larger than his substitution effect.
a) True
b) False

4. The game in the table below is an example of the Invisible Hand game.

		Me	
		Wash Hands	Don't Wash Hands
You	Wash Hands	3, 4	2, 1
	Don't Wash Hands	1, 2	1, 1

- a) True
b) False
5. The Nash equilibrium of a strategic interaction is always Pareto efficient.
a) True
b) False
6. When workers are paid their reservation wage, they are indifferent between keeping and losing their job.
a) True
b) False
7. If a price-taking firm sells one more unit of a good, then its marginal revenue will decrease.
a) True
b) False

8. Profit-maximising monopolies and firms in competitive markets produce a quantity at which marginal revenue equals marginal cost.
 - a) True
 - b) False
9. If the price falls in a market with price inelastic demand, the producer's total revenue will increase.
 - a) True
 - b) False
10. A farmer uses a pesticide on her wheat plantation to prevent pests from damaging the crop. However, the pesticide pollutes the river going through the farmer's land, which is used by a nearby community as a source of water. The marginal external cost is the cost of the pesticide to the farmer.
 - a) True
 - b) False

SECTION B

MULTIPLE CHOICE QUESTIONS

[30 marks: 2 marks for a correct answer and minus ¼ for an incorrect answer]

11. The GDP per capita of Greece was \$22 494 in 2012 and \$21 966 in 2013. Based on these figures, the growth rate of GDP between 2012 and 2013 (to two decimal places) was _____.
 - a) -2.40%
 - b) 2.35%
 - c) -2.35%
 - d) -0.24%
12. Which of the following statements regarding the Malthusian model are CORRECT?
 - I. A sustained increase in income per capita would be impossible.
 - II. The rise in productivity of labour would be offset by population growth.
 - III. Its predictions were correct throughout 19th century Britain.
 - IV. It explained why the population would stay constant over centuries without trending upwards.
 - a) I, II and IV are correct.
 - b) I and II are correct.
 - c) I and IV are correct.
 - d) I, II and III are correct.

13. Given a consumption choice model, assume that quantity of milkshakes is measured on the horizontal axis and the quantity of pizzas is measured on the vertical axis. At your optimal point of consumption an MRS of _____ means that 20 more pizzas are as good as 5 more milkshakes.

- a) $1/4$
- b) 0.2
- c) 20
- d) 4

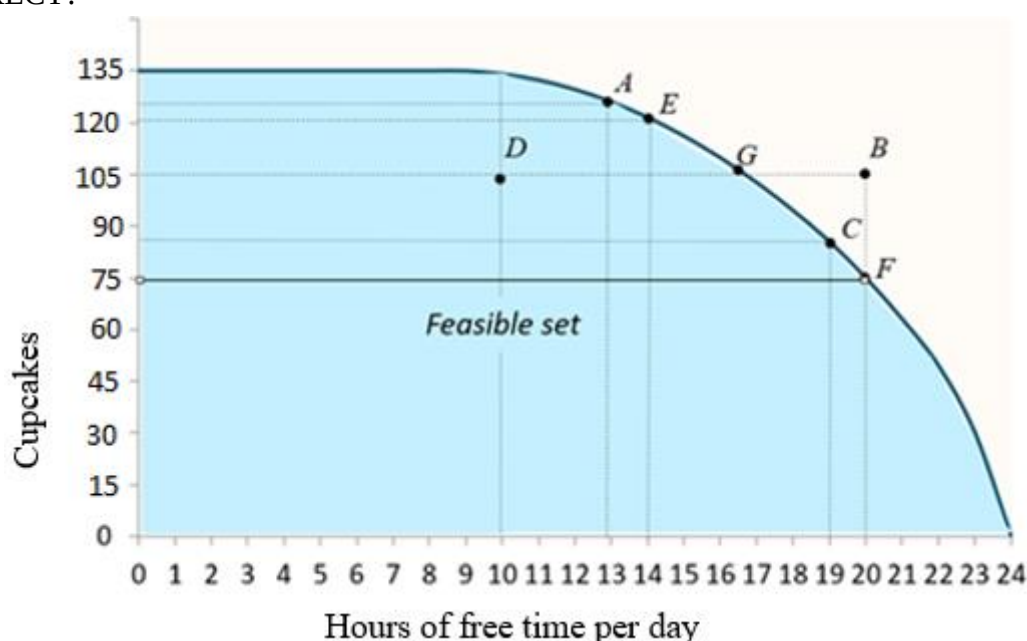
14. A worker faces a consumption constraint:

$$c = 10(24 - t)$$

where t is free time and c is consumption. Both free time and consumption are normal goods. If she was given an opportunity to receive R100 of unearned income a day, which one of the following statements applies to the worker?

- a) The worker will increase her consumption by R100.
- b) The worker will increase both her consumption and free time.
- c) The worker's budget line will pivot from the axis that measures her consumption.
- d) The worker's budget line will pivot from the axis that measures her hours of free time.

15. A baker's production of cupcakes corresponds to a certain number of hours worked per day. He faces the feasibility frontier below. Which ONE of the following statements are CORRECT?



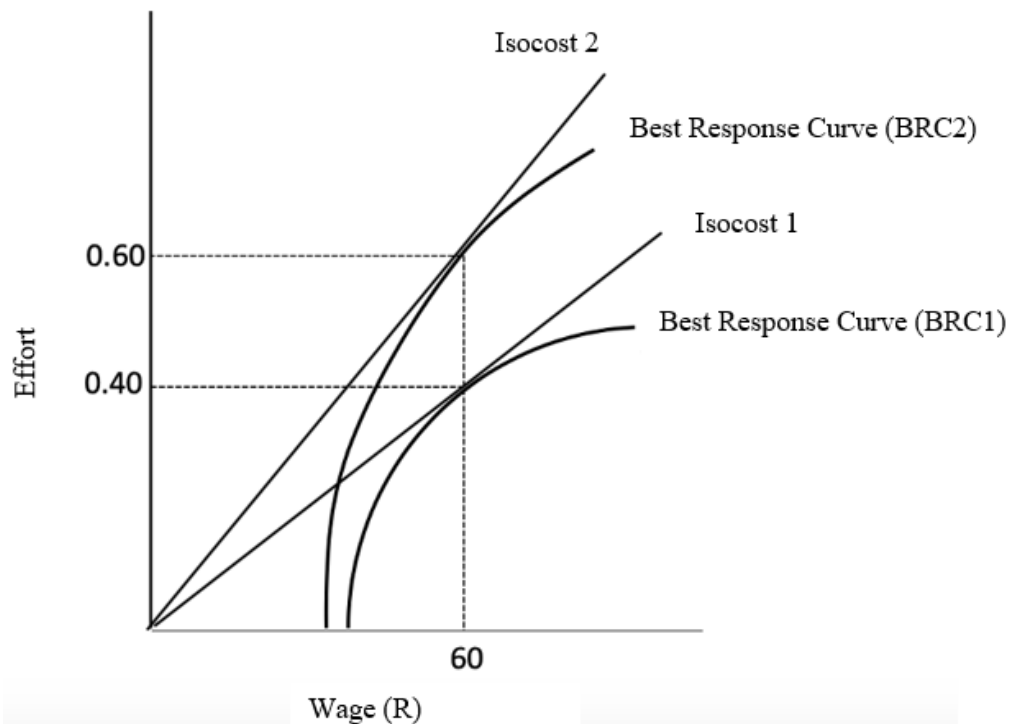
- a) Points B and D are both infeasible.
- b) Point D is not feasible because the baker is not taking as much leisure time as he could.
- c) The MRT between points E and F is approximately 7.5.
- d) At point C the baker's opportunity cost of free time is lower than at point E.

16. Which ONE of the following represents a social dilemma?
- Not everyone can agree whether VAT (value-added tax) should be increased.
 - Someone out for dinner with a group of friends decides to order the most expensive item on the menu after they all decided beforehand to share the total cost of the meal equally.
 - A good harvest of grapes is sold to neighbouring farms to produce wine.
 - Colleagues decide to carpool to work to save on the rising cost of petrol.
17. Zendaya and Tom Holland are having a staring contest, where the person who blinks first loses. If both of them blink, or if neither blinks, then there is a draw. Not blinking for too long dries their eyes out, so they would rather have a (Blink, *Blink*) draw than a (Don't blink, *Don't blink*) draw. The payoffs to their strategic interaction are represented in the figure below.

		Tom Holland	
		<i>Blink</i>	<i>Don't blink</i>
Zendaya	Blink	1, 1	0, 2
	Don't blink	2, 0	-1, -1

- Based on this information, which ONE of the following statements is INCORRECT?
- (Blink, *Don't blink*) is a Pareto-efficient outcome of this interaction.
 - (Don't blink, *Don't blink*) is Pareto dominated by all the Pareto-efficient outcomes of this interaction.
 - (Don't blink, *Don't blink*) is Pareto dominated by (Blink, *Don't blink*).
 - One Pareto improvement is possible on the outcome (Blink, *Don't blink*).
18. Which ONE of the following results in a higher reservation wage?
- Higher costs of searching for a job
 - The job is unpleasant.
 - Social stigma associated with unemployment
 - Lower unemployment benefits

19. A longer duration of expected unemployment shifts a worker's best response curve to the left from BRC1 to BRC2.



As a result of this change, which ONE of the following statements is CORRECT?

- a) The employer's cost of effort and the worker's employment rent have both increased
 - b) The employer should pay the worker less than R60 per hour to minimise costs.
 - c) The cost per unit of effort has decreased from R150 to R100.
 - d) The employer obtains fewer efficiency units per rand spent following the shift of the worker's best response curve.
20. Which of the following factors can shift the demand curve of a good?
- I. A change in consumer preferences.
 - II. A change in the price of the good.
 - III. A change in competitors' prices.
- a) II and III
 - b) I and III
 - c) Only I
 - d) Only II

21. Suppose the total cost function of a firm is given by:

$$TC = 150\,000 + 30Q + 3Q^2$$

The marginal cost of producing the 20th unit is:

- a) 147
- b) 150
- c) 6
- d) 210

22. If the absolute value of the price elasticity of demand for a product is more than one,
- a) a small price decrease will increase the firm's total revenue.
 - b) a small price decrease will decrease the firm's total revenue.
 - c) a small price increase will increase the firm's total revenue.
 - d) a small price increase has an undetermined effect on the firm's total revenue.
23. Which ONE of the following statements is TRUE regarding a firm that operates in a market where many firms produce homogeneous goods?
- a) The firm will exit the market if they make zero economic profit.
 - b) The average cost curve of the firm is the supply curve.
 - c) There are unexploited gains to trade if production occurs where marginal revenue is greater than marginal cost.
 - d) The firm will set its own price.
24. The education you receive at university has a positive effect on other members of society. Which ONE of the following statements is CORRECT?
- a) If nothing is done in an unregulated market for higher education, education will be undersupplied if universities are private enterprises.
 - b) A socially optimal provision of education could be achieved through taxing suppliers of education.
 - c) To ensure that the socially optimal quantity of education is provided, MSB must always exceed MPB.
 - d) Providing subsidies to the suppliers of education will decrease the quantity of education supplied.
25. Netflix, Disney Plus, Showmax and Amazon Prime are all streaming platforms, where people pay monthly to watch movies and TV series. Netflix is an example of:
- a) A rival and excludable good.
 - b) A rival and non-excludable good.
 - c) A non-rival and excludable good.
 - d) A non-rival and non-excludable good.

SECTION C
LONG QUESTIONS
[60 marks]

QUESTION 1

Briefly discuss why the United States and China would both be regarded as capitalist economies, using the definition of capitalism that was provided in this module. [2]

QUESTION 2

A firm, *Party Fits*, makes use of combinations of labour (L) and capital (K) to produce clothing. Suppose that the price of a unit of capital is R10. The firm can choose from three different technologies (A, B and C) to produce a particular quantity of clothing items. The firm currently uses technology A and the cost of technology A is R 850. The inputs of all three technologies are given below:

- A: 50 units of labour and 10 units of capital
- B: 25 units of labour and 40 units of capital
- C: 15 units of labour and 35 units of capital

- 2.1 How many units of capital does the firm need to give up to hire one additional worker? [1]
- 2.2 Given the relative price of labour, explain why *Party Fits* as a rational firm will not likely make use of a technology such as A. [3]
- 2.3 *Party Fits* competes with *Basic & Boujee* in the production of both shirts and trousers. The following table indicates the maximum amount of each clothing item that can be produced if the firms spend an hour on producing only that item. Both firms know that they can do better by specialising in the production of one of the goods. In which product should *Party Fits* specialise? [2]

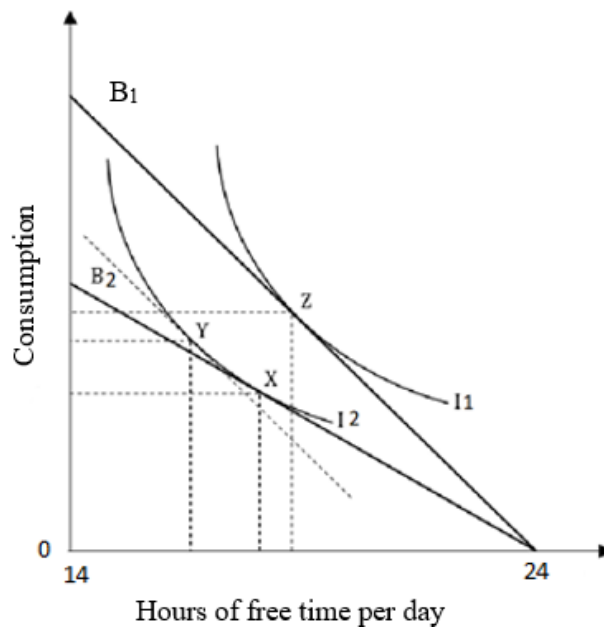
	Trousers	Shirts
Party Fits	12	5
Basic & Boujee	5	3

QUESTION 3

Leslie is hired as an employee by *Lovely Lawns*. She faces a linear consumption constraint, B_1 , represented by:

$$c = 200(24 - t)$$

where t is hours of free time and c is her consumption level.



3.1 What is Leslie's current hourly wage, and how many hours does she work per day? [1]

Suppose her wage is decreased by R100 an hour. Answer the following questions with reference to the graph above.

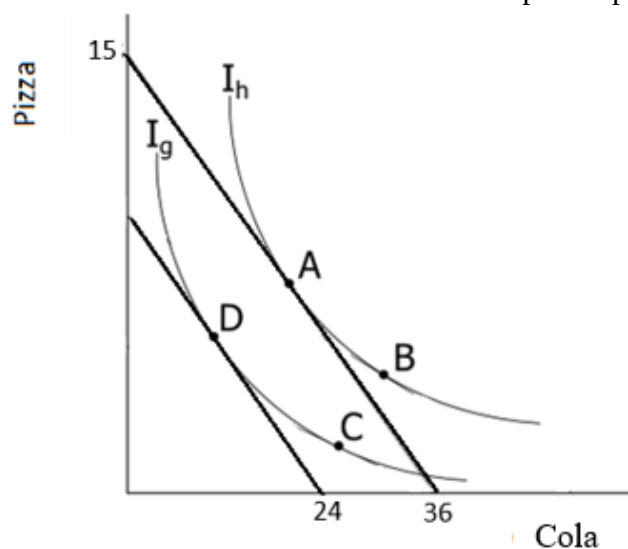
3.2 What is budget line B2's intercept on the axis for consumption? [1]

3.3 What is her MRS at her new optimal point? [1]

3.4 Is it safe to assume that a lower wage will encourage more hours worked for all individuals like it did for Leslie? Explain your answer. [2]

QUESTION 4

Carol is a consumer who earns an income of R360 per week. She uses her entire budget to purchase two goods: pizza and cola. Several indifference curves (I_g and I_h) and budget constraints are indicated in the figure below. Carol first consumes at her optimal point, **D**, but an increase in her income means that she consumes at a new optimal point, **A**.



4.1 What is the money price of a cola? [1]

4.2 What is the money price of a pizza?

[2]

4.3 What is her new income?

[1]

QUESTION 5

Consider the game below where two players, P1 and P2, have the option of playing either A or B.

		P2	
		A	B
P1	A	1, 8	2, 2
	B	7, 2	4, 3

5.1 What is the Nash equilibrium solution of this game?

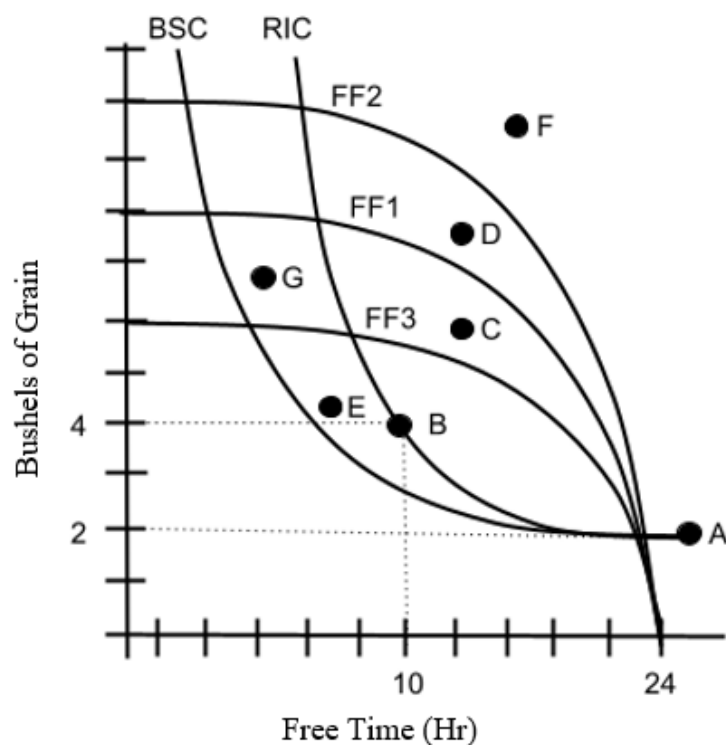
[2]

5.2 Does your answer to 5.1 also represent a dominant strategy equilibrium? Briefly explain.

[2]

QUESTION 6

Timon is a landowner and Pumbaa is a farmer who pays a share of his grain output to Timon for the use of the land. This is a voluntary exchange, where Timon makes Pumbaa a take-it-or-leave-it offer. Pumbaa's friend, Simba, agrees to give Pumbaa a survival ration of two bushels of grain if Timon's offer is rejected. The figure below shows several of Timon and Pumbaa's combined feasible frontiers (FF), Pumbaa's biological survival constraint (BSC) and reservation indifference curve (RIC). Timon and Pumbaa's current feasible frontier is FF1.



6.1 Which of the allocations – A, B, C, D, E, F, and G – are economically feasible? [2]

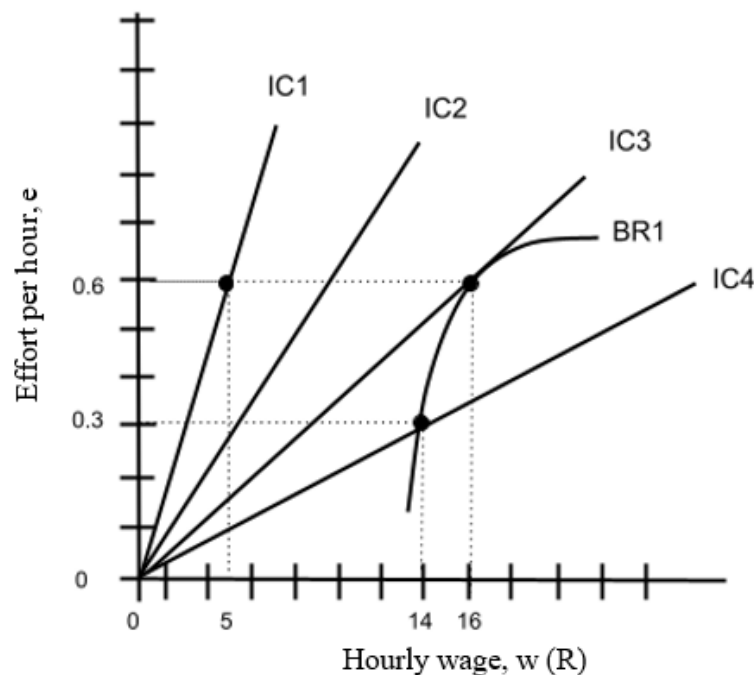
6.2 Would Pumbaa prefer allocation B to allocation A? [1]

Pumbaa's usual supplier of mulch has retired, and he has not been able to find a replacement. As a result, the acidity of the farm soil worsens, leading to worse grain harvests.

6.3 How does this affect the feasible frontier and Pumbaa's biological survival constraint? [1]

QUESTION 7

McDonald'sTM is hiring a new worker and is making use of the labour discipline model (given below), to set wages. The best response curve of this new worker is given by BR1. Round off all final answers to the second decimal place.



7.1 If McDonald'sTM could choose any isocost line on which to operate, which one of IC1, IC2, IC3 or IC4 would they choose and why? [2]

7.2 Which isocost lines are feasible? [1]

7.3 If McDonald'sTM sets an efficiency wage, what is the cost per unit of effort? [1]

QUESTION 8

Suppose the demand for wheat in Cofimvaba in the year 1825 is given by:

$$Q_d = 1000 - 2P$$

Favourable weather conditions cause the annual crop to increase from 350 to 550 tonnes. Calculate the price elasticity of demand over the quantity 350 to 550, and interpret your answer. Show all calculations. [5]

QUESTION 9

The demand curve of milk is given by the equation $P = 8 - Q_d$, whereas the supply curve is given by $P = \frac{1}{2}Q_s + 2$. The quantities Q_d and Q_s are measured in litres of milk.

- 9.1 Determine the equilibrium price and quantity algebraically. [3]
9.2 Calculate the situation that would arise in this competitive market if the price of milk is R5 per litre. How would the market respond? [3]

QUESTION 10

Fires in the Stellenbosch area have destroyed approximately 30% of vineyards this year. At the same time, the medical school of Stellenbosch University releases a statement to all people in Stellenbosch that a diet rich in grapes promotes intellectual performance. Explain with the aid of a graph the effect of these events on the grape market in Stellenbosch. [4]

QUESTION 11

The table below gives the demand and supply schedules for a competitive bread market.

<i>Price (Rands per loaf)</i>	<i>Quantity demanded (loaves per day)</i>	<i>Quantity supplied (loaves per day)</i>
6.00	240	0
8.00	200	80
10.00	160	160
12.00	120	240
14.00	80	320
16.00	40	400
18.00	0	480

- 11.1 What is the maximum price that consumers are willing to pay for the 80th loaf of bread? [1]
11.2 What is the minimum price that producers are willing to accept to produce the 200th loaf of bread? [1]
11.3 Calculate the consumer surplus if the market is in equilibrium. [2]

The demand for bread increases by 40 loaves of bread per day and the supply of bread decreases by 80 loaves of bread per day at each price.

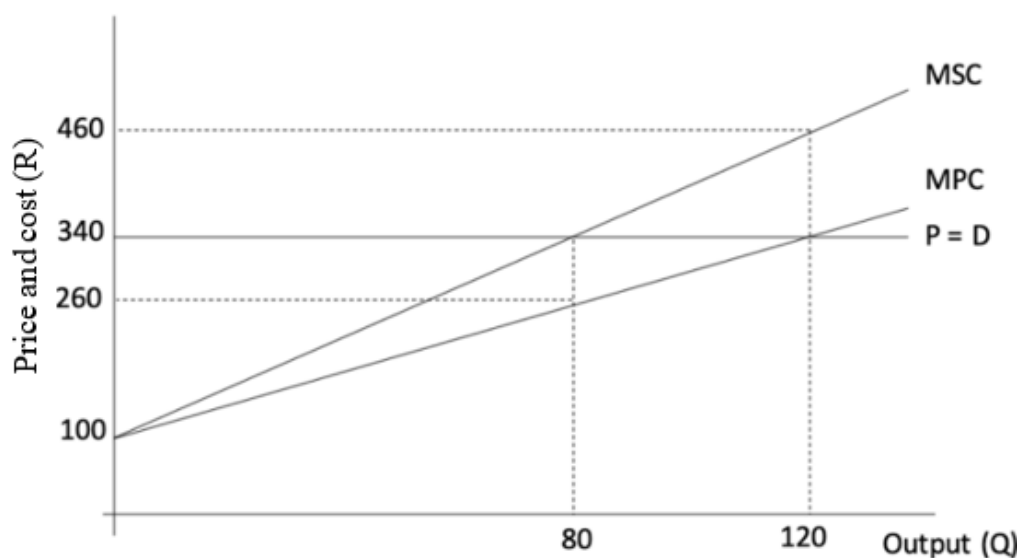
- 11.4 If the market is efficient, what is the new producer surplus? [4]

QUESTION 12

A unit tax is imposed on a good. Explain with the aid of a graph who carries the tax burden if supply is relatively elastic, but demand is perfectly inelastic. [4]

QUESTION 13

A factory producing cars is situated next to a university residence. The production process is noisy and disturbs the students when they attempt to study. The graph below depicts the MPC and MSC of car production. The car market is competitive and the market price is R340.



- 13.1 What is the socially efficient quantity of cars produced? [1]
13.2 Calculate the deadweight loss generated at the factory's profit-maximising output. [2]
13.3 What is the optimal per unit tax that could be charged by government to correct for the market failure? [1]

**UNIVERSITEIT VAN STELLENBOSCH
DEPARTEMENT EKONOMIE**

EKONOMIE 114

HOOFASSESSERING A3: 21 JUNIE 2022

Tyd: 3 ure

Punte: 100

EKSAMINATORE: Dr K Malindi Me L Boltman
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INTERNE MODERATOR: Dr E Moses

LET WEL: SLEGS NIE-PROGRAMMEERBARE SAKREKENAARS WORD TOEGELAAT.

BEANTWOORD **AFDELING A** VRAE OP DIE **ANTWOORDBLAD** WAT VERSKAF WORD. TOON U KEUSE DEUR 'N **KRUISIE** LANGS DIE NOMMER VAN DIE STELLING TE MAAK.

BEANTWOORD OOK **AFDELING B** VRAE OP DIE **ANTWOORDBLAD** WAT VERSKAF WORD. TOON U KEUSE DEUR 'N **KRUISIE** LANGS DIE NOMMER VAN DIE STELLING TE MAAK.

SLEUTEL:

AP = Gemiddelde produk
AFC = Gemiddelde vaste koste
AVC = Gemiddelde veranderlike koste
AC = Gemiddelde totale koste
BL = Begrotingslyn
BBP = Bruto Binnelandse Produk
IC = Indifferensie kurwe
K = Kapitaal
FF = Haalbare kurwe
MC = Marginale koste
MP = Marginale produk
MPB = Marginale private voordeel
MPC = Marginale private koste
MRS = Marginale koers van substitusie

MRT = Marginale koers van transformasie
MSB = Marginale sosiale voordeel
MSC = Marginale sosiale koste
P = Prys
PF = Produksiemoontlikheidskurwe
Q = Hoeveelheid
Q_d = Hoeveelheid gevra
Q_s = Hoeveelheid aangebied
TC = Totale koste
TP = Totale produk
w = Loon
WTA = Bereidheid om te aanvaar
WTP = Bereidheid om te betaal
Y = Inkome

AFDELING A
WAAR/ONWAAR STELLINGS

[10 punte: 1 punt vir 'n korrekte antwoord en minus ½ vir verkeerde antwoord]

1. BBP neem nie die waarde van huishoudelike of ander tipes onbetaalde werk in ag nie.
 - a) Waar
 - b) Onwaar
2. *Ceteris paribus* is 'n aanname in ekonomiese modelle wat verwys na die konstanthouding van die belangrike veranderlikes waarop jy fokus.
 - a) Waar
 - b) Onwaar
3. Indien Wendell sy hoeveelheid ure vrye tyd vermeerder ná 'n afname in sy uurlikse loon, was sy inkome-effek groter as sy substitusie-effek.
 - a) Waar
 - b) Onwaar

4. Die spel in die onderstaande tabel is 'n voorbeeld van die Onsigbare Hand spel.

		Ek	
		Was <u>Hande</u>	Was <u>Nie Hande</u>
Jy	Was <u>Hande</u>	3, 4	2, 1
	Was <u>Nie Hande</u>	1, 2	1, 1

- a) Waar
 - b) Onwaar
5. Die Nash ewewig van 'n strategiese interaksie is altyd Pareto-doeltreffend.
 - a) Waar
 - b) Onwaar
6. Wanneer werkers hul reserweloan betaal word, is hulle onsydig teenoor of hulle hul werk behou of verloor.
 - a) Waar
 - b) Onwaar
7. Indien 'n prysnemende firma een ekstra eenheid van 'n produk verkoop, sal sy marginale inkome verminder.
 - a) Waar
 - b) Onwaar

8. Winsmaksimerende monopolieë en firmas in mededingende markte produseer 'n hoeveelheid waar marginale inkome gelyk is aan marginale koste.
 - a) Waar
 - b) Onwaar
9. Indien die prys in 'n mark met prysonelastiese vraag daal, sal die produsente se totale inkome toeneem.
 - a) Waar
 - b) Onwaar
10. 'n Boer gebruik 'n plaagdoder op haar graanplantasie om te keer dat peste die graan vernietig. Die plaagdoder besoedel egter die rivier wat deur die boer se land loop en gebruik word as waterbron deur 'n nabygeleë gemeenskap. Die marginale eksterne koste is die koste van die plaagdoder vir die boer.
 - a) Waar
 - b) Onwaar

AFDELING B

VEELVULDIGE-KEUSE VRAE

[30 punte: 2 punte vir 'n korrekte antwoord en minus ¼ vir verkeerde antwoord]

11. Griekeland se BBP per capita was \$22 494 in 2012 en \$21 966 in 2013. Op grond van hierdie syfers was die groeikoers van BBP tussen 2012 en 2013 (tot twee desimale plekke)
 - a) -2.40%.
 - b) 2.35%.
 - c) -2.35%.
 - d) -0.24%.
12. Watter van die volgende stellings rakende die Malthusiaanse model is KORREK?
 - I. 'n Volgehoue toename in inkomste per capita is onmoontlik.
 - II. Die toename in arbeidsproduktiwiteit sal teengewerk word deur bevolkingsgroei.
 - III. Malthus se voorspellings was korrek vir Brittanje gedurende die 19de eeu.
 - IV. Dit het verduidelik waarom die bevolking konstant sou bly oor eeue sonder om opwaarts te neig.
 - a) I, II en IV is korrek.
 - b) I en II is korrek.
 - c) I en IV is korrek.
 - d) I, II en III is korrek.

13. Gegewe 'n verbruikskeuse model, neem aan dat die hoeveelheid melkskommels op die horisontale as en die hoeveelheid pizzas op die vertikale as gemeet word. By jou optimale verbruikspunt impliseer 'n MRS van _____ dat 20 meer pizzas net so goed soos nog 5 melkskommels is.

- a) $1/4$
- b) 0.2
- c) 20
- d) 4

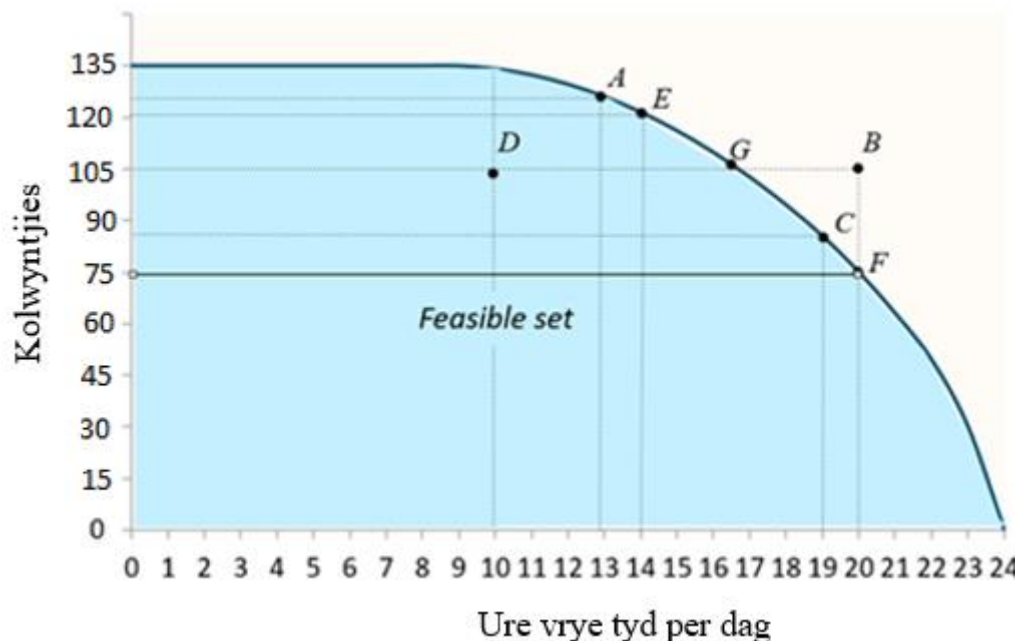
14. 'n Werker het 'n verbruiksbeperking van:

$$c = 10(24 - t)$$

waar t vrye tyd en c verbruik is. Beide vrye tyd en verbruik is normale produkte. Indien sy die geleentheid kry om R100 se onverdiende inkomste per dag te ontvang, watter een van die volgende stellings is van toepassing op die werker?

- a) Die werker sal haar verbruik met R100 verhoog.
- b) Die werker sal beide haar verbruik en vrye tyd verhoog.
- c) Die werker se begrotingslyn sal wegswaai van die as wat haar verbruik meet.
- d) Die werker se begrotingslyn sal wegswaai van die as wat haar ure vrye tyd meet.

15. 'n Bakker se produksie van kolwyntjies stem ooreen met die aantal ure wat hy per dag werk. Sy produksiemoontlikheidskurwe word hieronder getoon. Watter EEN van die volgende stellings is KORREK?



- a) Beide punte B en D is nie haalbaar nie.
- b) Punt D is nie haalbaar nie omdat die bakker nie soveel vrye tyd neem as wat hy kan nie.
- c) Die MRT tussen punte E en F is ongeveer 7.5.
- d) By punt C is die bakker se geleentheidskoste van vrye tyd minder as by punt E.

16. Watter EEN van die volgende stellings verteenwoordig 'n sosiale dilemma?

- a) Nie almal kan saamstem of BTW (belasting op toegevoegde waarde) verhoog moet word nie.
- b) Iemand wat saam met 'n groep vriende uiteet, besluit om die duurste item op die spyskaart te bestel nadat hulle almal vooraf besluit het om die totale koste van die ete gelykop te deel.
- c) 'n Goeie druiwe-oes word aan naburige plase verkoop om wyn te produseer.
- d) Kollegas besluit om saam werk toe te ry sodat hulle op stygende petrolkoste kan bespaar.

17. Zendaya en Tom Holland het 'n staarkompetisie waar die persoon wat eerste hulle oë knip, verloor. Indien beide van hulle hul oë knip of as nie een van hulle hul oë knip nie, dan is die uitkoms gelykop. As hulle vir te lank nie hulle oë knip nie, droog hulle oë uit en dus verkies hulle 'n (Knip, *Knip*) gelykop uitkoms bo 'n (Nie knip, *Nie knip*) gelykop uitkoms. Die betalings vir hul strategiese interaksie word in die figuur hieronder verteenwoordig:

		Tom Holland	
		<i>Knip</i>	<i>Nie knip</i>
Zendaya	Knip	1, 1	0, 2
	Nie knip	2, 0	-1, -1

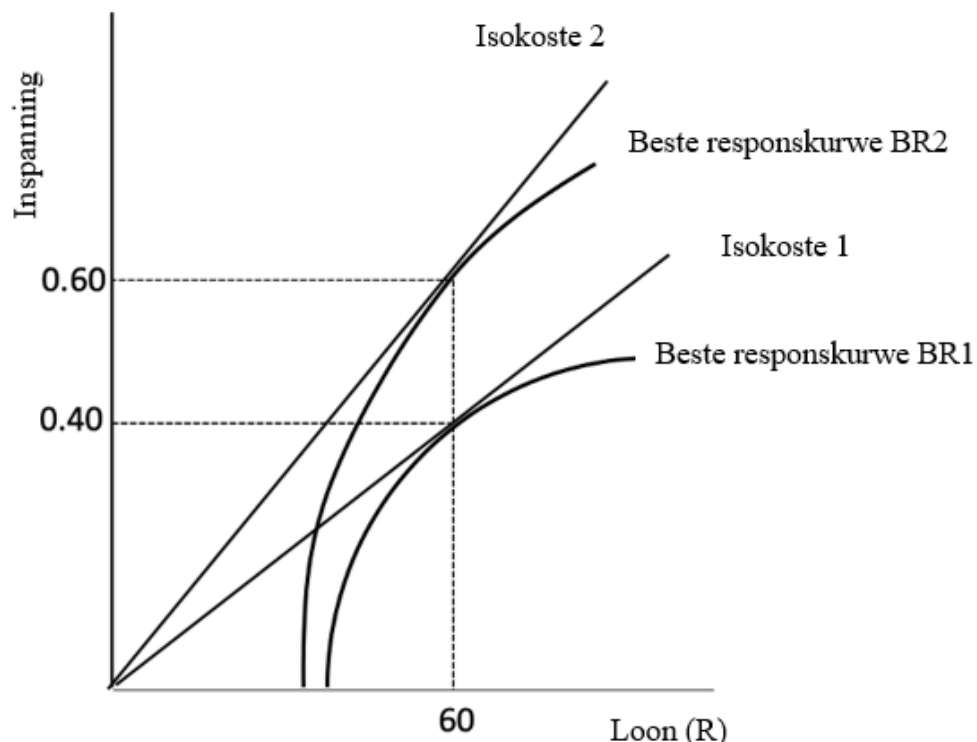
Gegewe hierdie inligting, watter EEN van die volgende stellings is VERKEERD?

- a) (Knip, *Nie knip*) is 'n Pareto-doeltreffende uitkoms van hierdie interaksie..
- b) (Nie knip, *Nie knip*) word Pareto-domineer deur al die Pareto-doeltreffende uitkomstes van hierdie interaksie.
- c) (Nie knip, *Nie knip*) word Pareto-domineer deur (Knip, *Nie knip*).
- d) Een Pareto verbetering is moontlik op die uitkoms (Knip, *Nie knip*).

18. Watter EEN van die volgende sal 'n verhoging in die reserweloan tot gevolg hê?

- a) Hoër kostes van werksoek
- b) Die werk is onaangenaam.
- c) Sosiale stigma verbonde aan werkloosheid
- d) Laer werkloosheidsvoordele

19. 'n Langer tydperk van verwagte werkloosheid skuif 'n werker se beste responskurwe linkswaarts van BR1 tot BR2.



As gevolg van hierdie verandering, watter EEN van die volgende stellings is KORREK?

- Die werkgewer se koste van inspanning en die werker se indiensnemings-opbrengste ("rent") het beide toeneem.
 - Die werkgewer moet die werker minder as R60 per uur betaal om koste te minimeer.
 - Die koste per eenheid van inspanning neem af vanaf R150 na R100.
 - Die werkgewer verkry minder doeltreffendheidseenhede per rand spandeer na die skuif van die werker se beste responskurwe.
20. Watter van die volgende faktore kan die vraagkurwe van 'n produk skuif?
- 'n Verandering in verbruikersvoorkeure.
 - 'n Verandering in die prys van die produk.
 - 'n Verandering in mededingers se pryse.
- II en III
 - I en III
 - Slegs I
 - Slegs II

21. Veronderstel die totale koste funksie van 'n firma word gegee deur:

$$TC = 150\,000 + 30Q + 3Q^2$$

Die marginale koste om die 20ste eenheid te vervaardig, is:

- a) 147
- b) 150
- c) 6
- d) 210

22. Indien die absolute waarde van pryselastisiteit van vraag van 'n produk groter as een is,

- a) sal 'n klein verlaging in die prys die firma se totale inkome verhoog.
- b) sal 'n klein verlaging in die prys die firma se totale inkome verlaag.
- c) sal 'n klein verhoging in die prys die firma se totale inkome verhoog.
- d) 'n klein verhoging in die prys sal 'n onbepaalde effek op die firma se totale inkome hê.

23. Watter EEN van die volgende stellings is WAAR vir 'n firma wat bedryf word in 'n mark waar baie firmas homogene produkte produseer?

- a) Die firma sal uit die mark tree indien 'n ekonomiese wins van nul gemaak word.
- b) Die gemiddelde kostekurwe van die firma is die aanbodkurwe.
- c) Daar is onbenutte voordele van handel by produksievlakke waar marginale inkome groter is as marginale koste.
- d) Die firma sal sy eie prys vasstel.

24. Die opvoeding wat jy op universiteit ontvang het, het 'n positiewe impak op ander lede van die gemeenskap. Watter EEN van die volgende stellings is WAAR?

- a) Indien niks in 'n ongeregleerde mark vir hoër onderwys gedoen word nie, sal opvoeding ondervoorsien word indien universiteite private instellings is.
- b) 'n Sosiaal optimale verskaffing van opvoeding kan verkry word deur die voorsieners daarvan te belas.
- c) Om te verseker dat die sosiale optimale hoeveelheid opvoeding verskaf word, moet MSB altyd die MPB oorskry.
- d) Deur subsidies aan die verskaffers van onderwys te betaal, sal die hoeveelheid opvoeding aangebied verlaag.

25. Netflix, Disney Plus, Showmax en Amazon Prime is almal “streaming” platforms waar mense maandeliks betaal om fliks en TV reekse te kyk. Netflix is 'n voorbeeld van:

- a) 'n mededingende en uitsluitbare produk.
- b) 'n mededingende en nie-uitsluitbare produk.
- c) 'n nie-mededingende en uitsluitbare produk.
- d) 'n nie-mededingende en nie-uitsluitbare produk.

AFDELING C
LANG VRAE
[60 punte]

VRAAG 1

Bespreek kortliks waarom die Verenigde State en China beide as kapitalistiese ekonomieë beskou sou word, deur gebruik te maak van die definisie van kapitalisme wat in hierdie module verskaf was. [2]

VRAAG 2

'n Firma, *Party Fits*, maak gebruik van 'n kombinasie van arbeid (L) en kapitaal (K) om klerasie te vervaardig. Veronderstel dat die prys van 'n eenheid kapitaal R10 is. Die firma kan kies tussen drie verskillende tegnologieë, A, B, en C, om 'n sekere hoeveelheid kledingstukke te vervaardig. Die firma gebruik tans tegnologie A en die koste van tegnologie A is R 850. Die insette van al drie tegnologieë word hieronder gegee:

A: 50 eenhede arbeid en 10 eenhede kapitaal

B: 25 eenhede arbeid en 40 eenhede kapitaal

C: 15 eenhede kapitaal en 35 eenhede kapitaal

2.1 Hoeveel eenhede kapitaal moet die firma prysgee om een addisionele werker aan te stel? [1]

2.2 Verduidelik waarom, gegewe die relatiewe prys van arbeid, 'n rasionele firma soos *Party Fits* waarskynlik nie 'n tegnologie soos A sal gebruik nie. [3]

2.3 *Party Fits* ding mee met *Basic & Boujee* in die produksie van beide hemde en broeke. Die volgende tabel dui die maksimum hoeveelheid van elke klere item aan wat geproduseer kan word indien die firmas 'n uur spandeer om slegs daardie produk te produseer. Beide firmas weet dat hulle beter kan doen deur te spesialiseer in die produksie van een van die goedere. In watter produk moet *Party Fits* spesialiseer? [2]

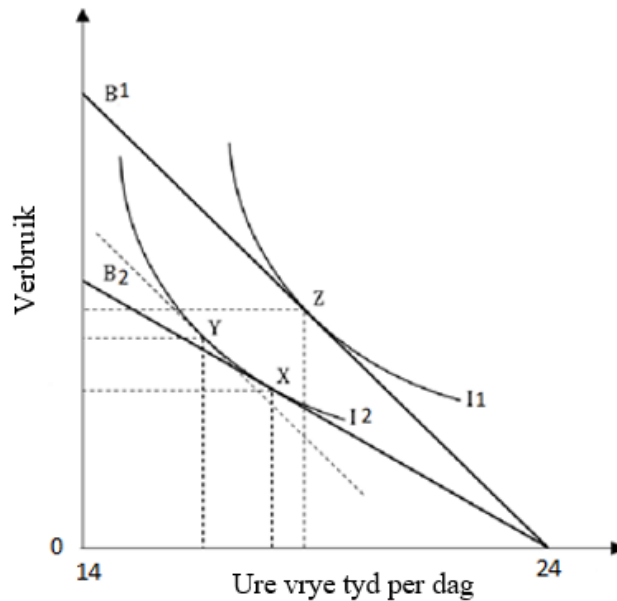
	Broeke	Hemde
Party Fits	12	5
Basic & Boujee	5	3

VRAAG 3

Leslie word gehuur as werknemer deur *Lieflike Tuine*. Sy het 'n lineêre verbruiksbeperking, B_1 , wat verteenwoordig word deur:

$$c = 200(24 - t)$$

waar t die aantal vrye tyd is en c haar verbruiksvlak.



3.1 Wat is Leslie se huidige uurlikse loon, en hoeveel ure werk sy per dag? [1]

Gestel haar loon word met R100 per uur verlaag. Beantwoord die volgende vrae met verwysing na die grafiek hierbo.

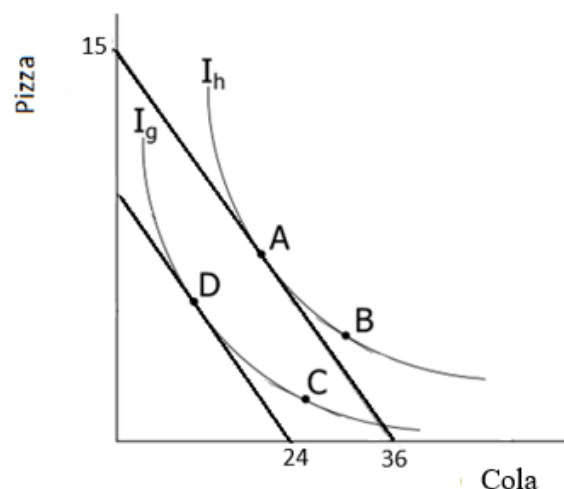
3.2 Wat is die waarde van begrotingslyn B₂ se afsnit op die as vir verbruik? [1]

3.3 Wat is haar MRS by haar nuwe optimale punt? [1]

3.4 Is dit veilig om aan te neem dat 'n laer loon alle individue sal aanmoedig om, soos Leslie, meer ure te werk? Verduidelik jou antwoord. [2]

VRAAG 4

Carol is 'n verbruiker wat 'n inkomste van R360 per week verdien. Sy gebruik haar hele begroting om twee produkte te koop: pizza en Cola. In die onderstaande figuur word verskeie indifferensiekurwes (I_g and I_h) en begrotingsbeperkings aangedui. Carol verbruik eers by haar optimale punt, **D**, maar dan neem haar inkomste toe en dan verbruik sy by haar nuwe optimale punt, **A**.



4.1 Wat is die geldprys van 'n Cola? [1]

4.2 Wat is die geldprys van 'n pizza? [2]

4.3 Wat is haar nuwe inkomste? [1]

VRAAG 5

Oorweeg die onderstaande spel waarin twee spelers, P1 en P2, die opsie het om of A of B te speel.

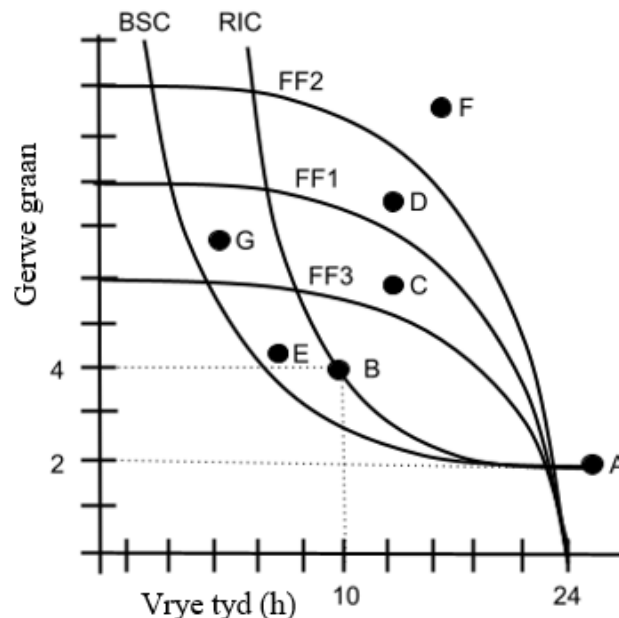
		P2	
		A	B
P1	A	1, 8	2, 2
	B	7, 2	4, 3

5.1 Wat is die Nash-ewewig oplossing vir hierdie spel? [2]

5.2 Verteenwoordig jou antwoord vir 5.1 ook 'n dominante strategie ewewig? Verduidelik kortliks. [2]

VRAAG 6

Timon is 'n grondeienaar en Pumbaa is 'n boer wat 'n deel van sy graanproduksie aan Timon betaal vir die gebruik van die grond. Dit is 'n vrywillige ruiltransaksie waar Timon aan Pumbaa 'n vat-dit-of-los-dit aanbod maak. Pumbaa se vriend Simba stem in om vir Pumbaa 'n oorlewingsrantsoen van twee gerwe graan te gee indien Timon se aanbod geweier word. Die figuur hieronder toon 'n aantal van Timon en Pumbaa se gesamentlike haalbare grense (FF), Pumbaa se biologiese oorlewingsbeperking (BSC) en sy reserwe indifferensiekurve (RIC). Timon en Pumbaa se huidige haalbare grens is FF1.



6.1 Watter van die allokasies – A, B, C, D, E, F, en G – is ekonomies haalbaar? [2]

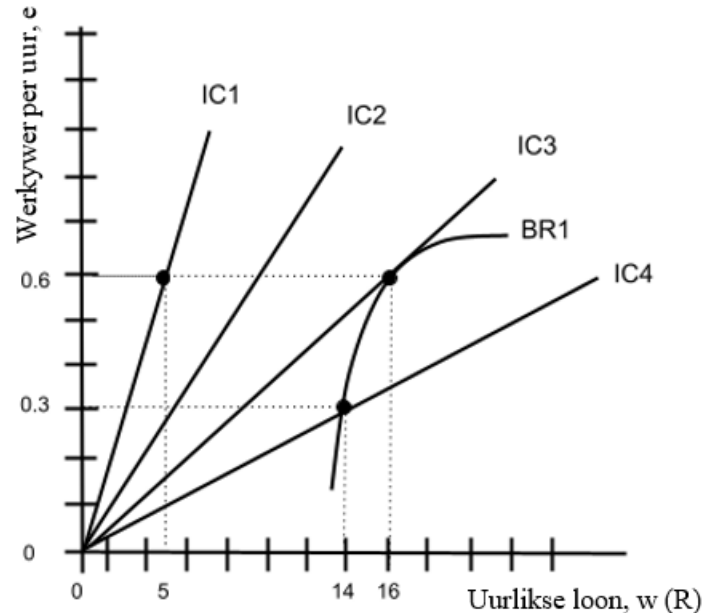
6.2 Sou Pumbaa allokasie B bo allokasie A verkies? [1]

Pumbaa se gewone verskaffer van deklaag het afgetree en hy was nog nie in staat om 'n plaasvervanger te kry nie. Gevolglik versleg die suurvlakke van die grond wat lei tot 'n kleiner graanoes.

6.3 Hoe beïnvloed dit die haalbare grens en ook Pumbaa se biologiese oorlewingsbeperking? [1]

VRAAG 7

McDonald'sTM stel 'n nuwe werker aan en maak gebruik van die arbeidsdissiplinemodel (wat hieronder gegee word) om lone vas te stel. Die beste reaksiekurwe van hierdie nuwe werker word deur BR1 gegee. Rond alle finale antwoorde tot die tweede desimaal af.



- 7.1 Indien McDonald'sTM enige isokoste lyn kon kies om op te werk, watter een van IC1, IC2, IC3 of IC4 sou hulle kies en hoekom? [2]
- 7.2 Watter isokoste lyne is haalbaar? [1]
- 7.3 Indien McDonald'sTM 'n doeltreffendheidsloon daarstel, wat is die koste per eenheid van inspanning (werksywer)? [1]

VRAAG 8

Gestel die vraag na koring in Cofimvaba in die jaar 1825 is:

$$Q_d = 1000 - 2P$$

Gunstige weersomstandighede veroorsaak dat die jaarlikse oes toeneem van 350 na 550 miljoen ton. Bereken die pryselastisiteit van vraag vir 'n verandering in hoeveelheid tussen 350 en 550 en interpreteer jou antwoord. Toon alle berekeninge. [5]

VRAAG 9

Die vraagkurwe van melk word gegee deur die vergelyking $P = 8 - Q_d$, terwyl die aanbodkurwe gegee word deur $P = \frac{1}{2}Q_s + 2$ is. Die hoeveelhede Q_d en Q_s word gemeet in liters melk.

- 9.1 Bepaal die ewewigsprys en -hoeveelheid algebraïes. [3]
- 9.2 Bereken die situasie wat sal ontstaan in hierdie mededingende mark indien die prys van melk R5 per liter is. Hoe gaan die mark reageer? [3]

VRAAG 10

Brande in die Stellenbosch omgewing het hierdie jaar sowat 30% van die wingerde vernietig. Terselfdertyd stel die mediese skool van die Universiteit van Stellenbosch 'n verklaring vry aan alle mense in Stellenbosch dat 'n dieët ryk aan druiwe intellektuele prestasie verbeter. Verduidelik met behulp van 'n grafiek die gevolge van hierdie gebeure op die druiwemark in Stellenbosch. [4]

VRAAG 11

Die onderstaande tabel toon die vraag-en aanbodskedules vir 'n mededingende broodmark.

Prys (Rande per brood)	Hoeveelheid gevra (brode per dag)	Hoeveelheid aangebied (brode per dag)
6.00	240	0
8.00	200	80
10.00	160	160
12.00	120	240
14.00	80	320
16.00	40	400
18.00	0	480

11.1 Wat is die maksimum prys wat verbruikers bereid is om te betaal vir die 80^{ste} brood? [1]

11.2 Wat is die minimum prys wat produsente bereid is om te aanvaar om die 200^{ste} brood te produseer? [1]

11.3 Bereken die verbruikersurplus as die mark in ewewig is. [2]

Die vraag na brood neem toe met 40 brode per dag en die aanbod van brood verminder met 80 brode per dag teen elke prys.

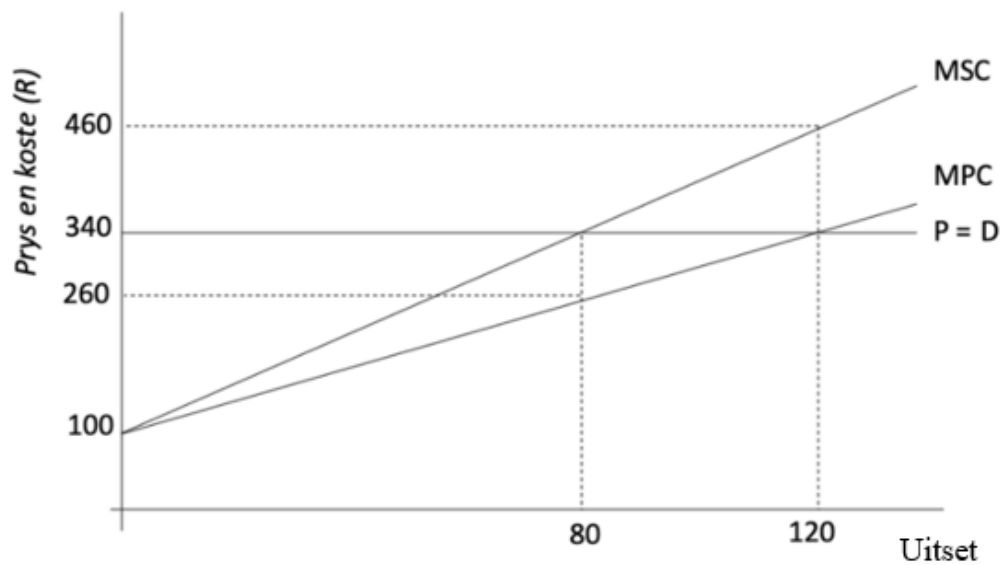
11.4 Indien die mark doeltreffend is, wat is die nuwe produsentesurplus? [4]

VRAAG 12

'n Eenheidsbelasting word ingestel op 'n produk. Verduidelik met behulp van 'n grafiek wie die belastinglas dra indien aanbod relatief elasties is, maar vraag volmaak onelasties is. [4]

VRAAG 13

'n Fabriek wat motors vervaardig is langs 'n studente koshuis geleë. Die produksieproses is raserig en steur die studente wanneer hul probeer studeer. Die onderstaande grafiek stel die MPC en MSC van motorproduksie voor. Die motormark is mededingend en die markprys is R340.



- 13.1 Wat is die sosiaal optimale hoeveelheid motors om te produseer? [1]
- 13.2 Bereken die dooiegewigsverlies wat by die fabriek se winsmaksimerende hoeveelheid motors veroorsaak word. [2]
- 13.3 Wat is die optimale belasting per eenheid wat deur die owerheid gehef kan word om die markmislukking reg te stel? [1]